

# ELTON CAO

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## EDUCATION

### Columbia University

New York, NY

*B.A., Computer Science and Mathematics, GPA: 3.96/4.0*

*May 2028*

- **Awards:** Science Research Fellow (top 0.1% of Columbia students). Dean's List (all semesters).
- **Regeneron STS T300 Scholar · JSHS State Champion · 3× ISEF Grand/Special Awards (\$12k total)**
- **Relevant coursework:** Data Structures & Algorithms, Systems Programming, ML for Genomics, Natural Language Processing, Discrete Mathematics, Computer Systems, Modern Algebra, High Performance ML

## PUBLICATIONS

- **Natural product based anticancer drug combination discovery assisted by deep learning and network analysis.** *Frontiers in Natural Products*, 2023. First author, 15 citations. [\[link\]](#)
- **National ground-level NO<sub>2</sub> predictions via satellite imagery driven convolutional neural networks.** *Frontiers in Environmental Science*, 2023. First author, 10 citations. [\[link\]](#)
- **Reconstruction of a 3D wireframe from a single line drawing via generative depth estimation.** Under review, *Computer-Aided Design*. First author. [\[link\]](#)

## EXPERIENCE

### Columbia University Creative Machines Lab

New York, NY

*Machine Learning Research Intern advised by Prof. Hod Lipson*

*Feb. 2025 – Present*

- Built a 638M-parameter ViT + diffusion model converting 2D engineering sketches into 3D.
- Engineered a GPU-accelerated pipeline to render >2 million CAD models, achieving a **4.5× speedup**.
- Implemented distributed training across an 8× H100 cluster, accelerating diffusion training by **3.2×**.
- Proposed a depth-based approach improving object coverage by **32%** over SOTA CAD reconstruction models.

### Amazon Web Services (AWS)

Santa Monica, CA

*Software Development Engineer Intern*

*Jun. 2026 – Aug. 2026*

- Led development of an automated Kubernetes application patching orchestrator for Managed Streaming Kafka (MSK) Connect, **accelerating patching processes by 5×**.
- Deployed RAG-assisted bug triaging workflows to surface patching errors, resulting in **2× faster** resolution times.

### National Oceanic and Atmospheric Administration (NOAA)

Boulder, CO

*Machine Learning Research Intern*

*Sep. 2021 – May 2024*

- Led design of a hybrid CNN-GNN framework in PyTorch for zero-shot nationwide coverage, achieving a **27% improvement** over SOTA NO<sub>2</sub> prediction models and solving significant scaling bottlenecks.
- Developed a parallelized system for **3× faster live data retrieval** via Google Earth Engine to process 5.9M satellite images, projected to **save \$100k+/year**.

### Leadspace

New York, NY

*Data Engineering Intern*

*May 2025 – Aug. 2025*

- Designed natural language enrichment algorithms yielding a **>40% improvement** in company feature identification and accuracy, **placing #1 categorically** and helping secure a **\$500k/year Salesforce contract**.
- Developed & deployed RAG-enhanced LangGraph agents in Databricks/PySpark across a 1M+ CRM dataset.

### University of Colorado Anschutz Medical Center

Denver, CO

*Machine Learning Research Intern*

*Oct. 2022 – Jan. 2024*

- Led investigation of synergy pathways in natural products, discovering **70+ candidate novel drug therapies**.
- Trained ensemble neural networks on molecular descriptors and drug-protein interactions on 10,000+ drugs.

## TECHNICAL SKILLS & INTERESTS

**Languages:** Python, C, C++, Go, Java, HTML/CSS, JavaScript, TypeScript, SQL, Bash, L<sup>A</sup>T<sub>E</sub>X

**Frameworks:** PyTorch, TensorFlow, Spark, React.js, Google Agent Dev Kit, LangChain/LangGraph

**Tools:** Databricks, AWS, Git, Docker, Google Cloud Platform, Google Earth Engine, Supabase, Linux, Kubernetes

**Libraries:** pandas, NumPy, Matplotlib, PySpark, scikit-learn

**Interests:** Basketball, Tennis, Squash, Pickleball, Acoustic/Electric Guitar & Bass